Hatcher - Algebraic Topology

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1 Basic definitions

We start by listing some very basic definitions necessary for the rest of the book.

1.1 Basic assumed topolgoical notions

Here are a few absolutely elementary topological definitions, that are written down for good measure.

The first is the product topology. It is easy to remember what it is, just by thinking about the categorical definition of a product (as an inverse limit).

Definition 1.1.1: Product topology

Let $\{X_i\}$ be a family of spaces, and let $X := \prod_i X_i$ be the cartesian product of these spaces. The **product topology** on X is the coarsest topology on X such that all the projection maps $p_i : X \to X_i$ are continuous. Equivalently, it is given by the basis $\{p_i^{-1}(U_{ij})\}$, where U_{ij} runs over the open sets $U_{ij} \subseteq X_i$. Again, in other words, the open sets of X is given by the (possibly infinite) union of sets of the form $\prod_i U_i$, where $U_i \neq X_i$ for only finitely many i.

Similarily, the definition of a quotient topology, is pretty much the only thing it can be.

Definition 1.1.2: Quotient topology

Let X be a set, and $\sim$ an equivalence relation on X. Define $Y := X/\sim$, and let $p : X \to Y$ be the canonical projection. The **quotient topology** on Y is defined such that $U \subseteq Y$ is open if and only if $p^{-1}(U)$ is open.

There are more categorical ways to define the quotient topolology as well: For any surjection $q : X \to Y$, we can define the quotient topology on Y as the **final topology** wrt. q: this is the finest topology on Y, that makes q continuous (in general, the final topology is essentially the topology that appears when taking direct limits). It has the universal property that

$$f : Y \to Z \text{ continuous} \Leftrightarrow f \circ g : X \to Z \text{ continuous}$$

The dual notion of the final topology is the initial topology, of which the product topology was an example (not surprising, since that would be an inverse limit).

We also recall that a space is said to be **connected** if it is not possible to write as a disjoint union of two or more non-empty open subsets. A stronger notion is that of a **path-connected** space, in which every point has a path connecting them. A space is **locally path-connected** if every point $x \in X$ admits arbitrarily small neighborhoods (i.e. for every open $x \in U$, there exists a $x \in V \subset U$) that are path-connected. Both path connected and the locally path connected assumption may be necessary, since they actually don't imply each other (but the spaces that are path-connected but not locally path connected are pretty wacky).

Retractions, Deformation retracts.

2 The Fundamental Group

2.1 Basic constructions

We recall the definition of the first invariant of a space we study, the fundamental group.

Definition 2.1.1: The Fundamental Group

Let (X, x_0) be a pointed space. The fundamental group $\pi_1(X, x_0)$ is defined as the homotopy classes $[f]$ of loops in x_0, under composition of paths.

Proposition 2.1.2

Let h be a path in X from x_0 to x_1. Then h induces an isomorphism

$$\beta_h : \pi_1(X, x_0) \to \pi_1(X, x_1).$$

Proof. Just do the only thing you can do: Compose loops with h to and from. Everything then follows again since we only care about paths up to homotopy. □

Because of this, we often omit the base point x_0 from notation when X is path-connected.

By the above, the following definition is independent of base point. The following definition can also be described to say that every loop in X is contractible (nullhomotopic).

Definition 2.1.3

A path-connected space X is said to be **simply connected** if $\pi_1(X, x_0)$ is trivial for one choice (and hence all choices) of base-point x_0.

The first goal of this chapter is then to prove $\pi_1(S_1) \cong \mathbb{Z}$. This is done by introducing the quite important and general notion of a covering space.

Definition 2.1.4: Covering Space

A **covering space** $(\tilde{X}, p)$ of X is a space $\tilde{X}$ and a map $p : \tilde{X} \to X$ such that every point $x \in X$ has an open neighborhood $U_x \subset X$ such that $p^{-1}(U_x)$ is the disjoint union $\sqcup V_x$ of spaces V_x that are all mapped homeomorphically onto U_x by p. Moreover, such an U_x is said to be **evenly covered**.

The first example of a covering space, which will be used to compute the fundamental group of S^1 is the map

$$\begin{aligned} p : \mathbb{R} &\to S^1 \\ r &\to (\cos 2\pi r, \sin 2\pi r). \end{aligned}$$

Covering spaces are discussed in more detail in Section 2.3. Their lifting properties and similar will be discussed there. Assuming these lifting properties, the book then goes on to prove the structure of $\pi_1(S^1)$, and then prove numerous cool consequences.

It is clear that the construction of π_1 is in fact functorial: continuous maps between pointed spaces $s : (X, x_0) \to (Y, y_0)$ clearly induces homomorphisms $\varphi_s : \pi_1(X, x_0) \to \pi_1(Y, y_0)$, and identities and associativity conditions are trivial.

The following will imply the easy part of Van Kampen's theorem, but we state it here, as the book.

Lemma 2.1.5

Let X be a space, and let $\{A_\alpha\}_\alpha$ be path-connected open sets that cover X, with each open containing the basepoint $x_0 \in A_\alpha$. If each intersections $A_\alpha \cap A_\beta$ are all path-connected, then each loop in X based at x_0 is homotopic to a product of loops, all contained in a single A_α.

Proof. The idea is very simple: a loop f can be partitioned into pieces all contained in a single A_α. Between each partition, we are in an intersection, and we can then go back and forth between the base-point x_0. □

A nice consequence is already the following

Proposition 2.1.6

$\pi_1(S^n) = 0$ for $n \geq 2$.

Proof. Let A_1, A_2 be the compliments of two antipodal points on S^n. Then $A_i \cong \mathbb{R}^n$, and $A_1 \cap A_2 \cong \mathbb{R} \times S^{n-1}$, which is path-connected for $n \geq 2$. Thus, for any basepoint $x_0 \in A_1 \cap A_2$, Lemma 2.1.5 implies that every loop f is homotopic to a product of loops in A_1 and A_2 separately, but $\pi_1(A_i) = 0$, since they are both homeomorphic to $\mathbb{R}^n$, and hence f must be nullhomotopic. □

2.2 Van Kampen

The Van Kampen theorem is probably the first major theorem in the book, and allows us to compute fundamental groups from sufficiently nice covers.

Theorem 2.2.1: Van Kampen

Let X and A_α be as in Lemma 2.1.5. If each intersection $A_\alpha \cap A_\beta$ is path-connected, then the homomorphism $\Phi : *_\alpha \pi(A_\alpha, x_0) \to \pi(X, x_0)$ is surjective. If additionally all the triple intersections $A_\alpha \cap A_\beta \cap A_\gamma$ is path-connected, then the kernel of Φ is generated by the elements of the form $\iota_{\alpha\beta}(\omega)\iota_{\beta\alpha}(\omega)^{-1}$, where $\iota_{\alpha\beta}$ denotes the homomorphism $\pi_1(A_\alpha \cap A_\beta, x_0) \to \pi_1(A_\alpha)$ induced by the inclusion.
Specializing to the case where there are only two open sets A_1, A_2, the theorem states that when $A_1 \cap A_2$ is path-connected, $\pi_1(X, x_0)$ is exactly the pushout (in the category of groups)

$$\begin{array}{ccc} \pi_1(A_1 \cap A_2, x_0) & \longrightarrow & \pi_1(A_2, x_0) \\ \downarrow & & \downarrow \\ \pi_1(A_1, x_0) & \longrightarrow & \pi_1(X, x_0) = \pi_1(A_1, x_0) *_{\pi_1(A_1 \cap A_2, x_0)} \pi_1(A_2, x_0) \end{array}$$

Proof. It is easy to see that Φ is a homomorphism (induced by the inclusions $A_\alpha \hookrightarrow X$). Then, the surjectivity part is just Lemma 2.1.5.

To describe the kernel, we can choose a "factorization" of a loop $[f] \in \pi_1(X, x_0)$, (i.e. write f as a composition of loops all contained in single A_α's), and then show that all such choices give the same element once we quotient out by the relations describing the kernel of Φ. The idea of the proof of this is to consider a homotopy $I \times I \to X$ between two factorizations, and subdivide this square into a bunch of rectangles, each contained in a single A_α. Then one sees that moving across the rectangles may only change the factorizations by one of the relations. □

There are many good applications of Van Kampen in the book. A particularly nice one is the computation of the fundamental group of a cell-complex.

Proposition 2.2.2

Let Y be a space obtained from X by attaching 2-cells e^2_α along $\varphi_\alpha : S^1 \to X$. The loops defined by φ_α (this involves choosing a path to the basepoint x_0 to any point in this loop) are all clearly null-homotopic in Y, and thus lie in the kernel of the map $\varphi_i : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ induced by the inclusion. Denote the normal subgroup generated by all these loops by N. Then:

- φ_i is surjective with kernel N.
- If Y is instead obtained from X by attacking n-cells with $n > 2$, then φ_i is an isomorphism.
- For a path connected cell-complex, the inclusion of the 2-skeleton $X^2 \hookrightarrow X$ induces an isomorphism $\pi_1(X^2, x_0) \to \pi_2(X, x_0)$.

Proof. The proof of a) is quite nice and visual, see the book. Proof of b) is the same as a), but with a different conclusion. And finally c) follows easily from b) (at least in the finite-dimensional case,

but the infinite dimensional case need to take a bit more care, but is still true). □

One nice corollary of the proposition above is essentially a solution to the topological version of the inverse Galois problem (the relation between galois theory and the fundamental group becomes a bit more clear in the next section, and rigorous in a later note on Étale cohomology).

Corollary 2.2.3

Let G be a group. There exists a 2-dimensional cell complex X_G with $\pi_1(X_G) \cong G$.

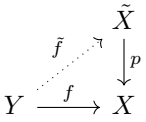
Proof. Write G from its finite presentation $\langle g_\alpha \mid r_\beta \rangle$ (i.e. g_α being the free generators, and r_β the words defining the relations). Then form X_G by starting with $\bigvee_\alpha S_\alpha$, and attaching 2-cells e^2_β by the loops defined by the relations r_β. □

2.3 Covering spaces

In this section, we describe covering spaces, whose definition was given eawrlier, in much more detail. The main goal of the section is to show the Galois-correspondence between covering spaces and the fundamental group.

Throughout this section, for a space X, $\tilde{X}$ will denote a covering space, and $p : \tilde{X} \to X$ will denote the corresponding covering map.

The raison d'etre of covering spaces is their lifting properties. Recall that given a map $f : Y \to X$, a lift of f is a map $\tilde{f}$ such that the following diagram commutes



The first lifting property is the **homotopy lifting property**.

Proposition 2.3.1: Homotopy lifting property

Given a homotopy $f_t : Y \to X$, and a map $\tilde{f}_0$ lifting f_0, there exists a unique homotopy $\tilde{f}_t : Y \to \tilde{X}$ of $\tilde{f}_0$ lifting f_t. In particular, taking Y to be a point, we get the **path lifting property**, that for every path $f : I \to X$, and choice of lift $\tilde{x_0}$ of $x_0 = f(0)$, there exists a unique path $\tilde{f} : I \to \tilde{X}$.

Proof. The proof is done in three steps: For a given point $y \in Y$, we can construct a neighborhood N and a lift of $F : N \times I \to X$. This is done by segmenting I into segments that are evenly covered. By compactness of I, this requires only finitely many segments.

Next, one shows that such a lift is unique for each point (this is direct from the definition of a covering space), and thus the lifts constructed in the first step glue together. □

There is immidiately a nice application of the homotopy lifting property in the following

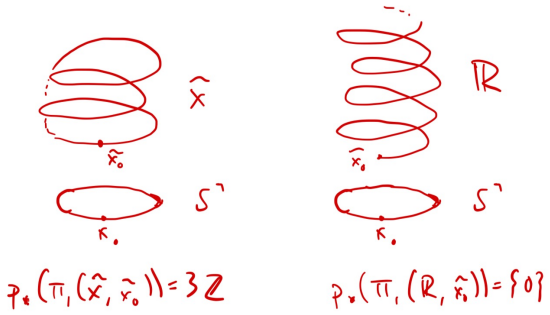
Proposition 2.3.2

The map $p_* : \pi_1(\tilde{X}, \tilde{x_0}) \to \pi_1(X, x_0)$ induced by a covering space is injective. The image $p_*(\pi_1(\tilde{X}, \tilde{x_0}))$ consists of homotopy classes of loops starting in x_0 whose lifts become loops in $\tilde{X}$ starting in $\tilde{x_0}$.
Assuming $X, \tilde{X}$ are both path-connected, the number of sheets of the covering space $p : (\tilde{X}, \tilde{x_0}) \to (X, x_0)$ is equal to the index of $p_*(\pi_1(\tilde{X}, \tilde{x_0}))$ in $\pi_1(X, x_0)$.

Proof. By the homotopy lifting property, any loop that is null-homotopic in (X, x_0) would also give rise to a null-homotopy of the original loop. For the second statement, it is clear that such loops are in the image. However, any loop in $p_*(\pi_1(\tilde{X}, \tilde{x_0}))$ must be at least homotopic to such a loop, but by homotopy-lifting, it must also lift to a loop starting in $\tilde{x_0}$.

For the last statement, let $H = p_*(\pi_1(\tilde{X}, \tilde{x_0}))$, and notice that there is a well-defined function from the cosets of H to $p^{-1}(x_0)$, by the previous paragraph, sending a coset $H[g]$ to $g(1)$. This function is clearly injective, as if $g_1(1) = g_2(1)$, then we have $g_1\overline{g_2} \in H$, and it is surjective by the path-connectedness of $\tilde{X}$. □

The image of a covering space can be intuitively visualized by the following pretty ugly-looking drawing. It is also easy to see the statement about the sheets.



We now state the next two important lifting properties,

Proposition 2.3.3: The lifting criterion

Given a map $f : (Y, y_0) \to (X, x_0)$, where Y is path-connected and locally path-connected, a lift $\tilde{f} : (Y, y_0) \to (\tilde{X}, \tilde{x_0})$ exists if and only if $f_*(\pi_1(Y, y_0)) \subseteq p_*(\tilde{X}, \tilde{x_0})$.

Proof. The only if part is obvious, since $p\tilde{f} = f$. The idea for the other direction, is to define $\tilde{f}(y)$ by taking a path α from y_0 to y, and lifting $f\alpha$ and taking the end-point. The assumption is then necessary to get well-definedness. □

The final property is the following

Proposition 2.3.4: Unique lifting property

Assume Y is connected. If two lifts $\tilde{f}_1, \tilde{f}_2 : Y \to \tilde{X}$ of a map $f : Y \to X$ agree on a single point, then they agree on all of Y.

Proof. The idea is to show that both the set of points that they do and don't agree must be open. Thus The set of points where they agree is both open and closed, and connectedness then implies it must be either empty or all of Y. □

Next, we move on to the classification of covering spaces, which will bear a strong resemblence to Galois theory, or even more so to class-field theory.

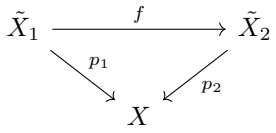
First, we need the following definition

Definition 2.3.5: semi-locally simply connected

A space X is said to be **semi-locally simply connected** if for every point $x \in X$, there exists an open $x \in U$ such that the inclusion induced map $\pi_1(U, x) \to \pi_1(X, x)$ is trivial.

Note that in the above definition $\pi_1(U, x)$ need not itself be trivial. What the definition says is that loops in U at x are contractible in X, but the contraction need not all happen inside of U.

Next, when talking about isomorphisms of covering spaces, we of course mean a map $f_1 : \tilde{X}_1 \to \tilde{X}_2$ such that the following commutes:



This allows us to state

Theorem 2.3.6: Classification of covering spaces

Let X be path-connected, locally path-connected and semi-locally simply connected. Sending a path-connected covering space $p : \pi_1(\tilde{X}, \tilde{x_0})$ to the subgroup $p_*(\pi_1(\tilde{X}, \tilde{x_0})) \subset \pi_1(X, x_0)$ induces a bijection between isomorphism-classes of base-point preserving path-connected covering-spaces and subgroups of $\pi_1(X, x_0)$.
Ignoring base-points, the association induces a bijection between path-connected covering spaces, and conjugacy classes of subgroups of $\pi_1(X, x_0)$.

Proof. The main idea is to construct a simply-connected covering space just from X and $\pi_1(X, x_0)$. This space is constructed as

$$\tilde{X} := \{[\gamma] \mid \gamma \text{ a path starting in } x_0\},$$

with a topology defined as follows: First, let $\mathcal{U}$ be the collection of path-connected open sets such that $\pi_1(U) \to \pi_1(X)$ is trivial. The assumptions means that this is a basis of the topology on X. For each $U \in \mathcal{U}$, we can define open sets of the form

$$U_{[\gamma]} := \{[\gamma\eta] \mid \eta \text{ is a path in } U \text{ with } \gamma(1) = \eta(0)\}.$$

This forms a basis of a topology on $\tilde{X}$. From this, most of the stuff follows surprisingly smoothly by using the definitions (see pages 63 - 68 in Hatcher). □

Of course, the above implies the existence of a unique (up to isomorphism) path-connected cover of X corresponding to the trivial subgroup (that would be the simply connected cover that was constructed in the proof). This is called the **universal cover** of X.

Next, we go on to describe deck-transformations. Under composition, these form a group, which is really the analogy of Galois groups.

Definition 2.3.7: Deck transformation

An automorphism of a covering space $\tilde{X} \to X$ is called a **deck transformation**. The group of all deck transformations under composition is denoted $G(\tilde{X}/X)$.

We need another definition whose name will be immidiately justified.

Definition 2.3.8: Normal covering (also called Regularing covering)

A covering space $p : \tilde{X} \to X$ is called **normal** (or sometimes **regular**) if for each $x \in X$ and each pair of lifts $\tilde{x}_1, \tilde{x_2}$ of x, there exists a deck transformation taking $\tilde{x_1}$ to $\tilde{x}_2$.

Proposition 2.3.9

Let X be path-connected and locally path-connected, and let H be the subgroup of $\pi_1(X, x_0)$ corresponding to a path-connected covering space $\tilde{X}$. Then

- The covering is normal if and only if $H < \pi_1(X, x_0)$ is normal.
- We have $G(\tilde{X}/X) \cong N(H)/H$, where $N(H)$ is the normalizer of H.

In particular, the universal cover of X satisfies $G(\tilde{X}/X) = \pi_1(X, x_0)$.

Proof. It is easy to see that changing basepoints of a cover correspond to conjugating H by an element of $\pi_1(X, x_0)$ (one that lifts to a path between the two choices of basepoint of $\tilde{X}$). Thus, an element $[\gamma] \in \pi_1(X, x_0)$ (which lifts to a path from $\tilde{x_0}$ to $\tilde{x_1}$) is in the normalizer of H if and only if $p_*(\pi_1(\tilde{X}, \tilde{x_0})) = p_*(\pi_1(\tilde{X}, \tilde{x_1}))$. Now apply the lifting criterion to see that this condition is equivalent to the existence of a deck transformation from that takes $\tilde{x_0}$ to $\tilde{x_1}$. The first point follows.

The second point is then obvious, by considering the map $N(H) \to G(\tilde{X}/X)$ given by sending a an element of the normalizer of H to the corresponding deck-transformation. Of course, elements of H act trivially, since they lift to loops. □

A lot of this material on covering spaces is analogous to Galois theory. We can think of a cover as an extension of its base space, and of deck transformations as symmetries that fix the base. The following table gives an informal dictionary between covering space theory and Galois theory. Pushing this idea to the fullest leads to the **Étale fundamental group**, which unifies these two viewpoints. Maybe in a later note, we can dive into this topic, but for now the following table will do:

Covering space theory	**Galois theory**
The base space X.	The base field K.
A connected covering space $\tilde{X} \to X$.	A separable algebraic field extension L/K.
The number of sheets of a finite cover: how many points lie over a point of X.	The degree $[L : K]$ of a finite extension: how large the extension is over K.
The universal cover $U \to X$: a simply connected space covering every connected cover of X.	The separable closure K^{sep}/K: a field containing a copy of every separable algebraic extension of K.
The fundamental group $\pi_1(X, x_0) \cong G(U/X)$ controls the covering spaces of X.	The absolute Galois group $\text{Gal}(K^{\text{sep}}/K)$ controls the separable extensions of K.
The deck group $G(\tilde{X}/X)$: symmetries of the cover that leave its projection to X unchanged.	The automorphism group $\text{Aut}(L/K)$: symmetries of L that fix every element of K.
A normal (regular) cover: its deck transformations can move any point of a fibre to any other.	A Galois extension: all the conjugates stay inside the extension, giving the full collection of symmetries.
A cover corresponds to the subgroup $H = p_*(\pi_1(\tilde{X}, \tilde{x}_0))$. Its elements are the loops whose lifts close up.	An intermediate field L corresponds to $\text{Gal}(K^{\text{sep}}/L)$, the automorphisms that fix L pointwise.
Smaller subgroups give larger covers, with more sheets. Normal subgroups give normal covers.	Smaller fixing subgroups give larger fields. Normal subgroups give Galois extensions.
An intermediate cover gives a tower $\tilde{X} \to Y \to X$.	An intermediate field gives a tower $K \subseteq F \subseteq L$. Notice the reversal of arrows.
Choosing $\tilde{x}_0$ in the fibre picks out an actual subgroup; changing this choice conjugates the subgroup.	Choosing an embedding $L \hookrightarrow K^{\text{sep}}$ picks out a particular copy of L; changing it conjugates the fixing subgroup.
Lifting loops permutes the sheets of a cover.	Galois automorphisms permute the embeddings of an extension, or the roots of a defining polynomial.

3 Homology

In this chapter, we define and discuss the basic properties of simplicial and singular homology. We can be quite quick in the discussion of abstract machinery, since they are covered in other notes, but arguably, these homology theories are the simplest to understand, and thus their definition is nice for intuition (even if, as Hatcher nicely puts it, the properties are the most important points for working with it in practice).

3.1 Simplicial and Singular Homology

Recall the following

Definition 3.1.1: n-simplex, faces, boundaries

An **n-simplex** is defined as

$$\Delta^n := \{(t_0, \dots, t_n) \in \mathbb{R}^{n+1} \mid \sum_i t_i = 1, t_i \geq 0\},$$

thought of as an n-dimensional triangle, together with an ordering on the vertices $[v_0, \dots, v_n]$, according to the index (this gives an orientation of each edge $[v_i, v_j]$). A **face** of Δ^n, is the Δ^{n-1} obtained by deleting one of the verteces, written $[v_0, \dots, \hat{v_i}, \dots, v_n]$. The boundary $\delta\Delta^n$ is the union of all the faces, while the **open n-simplex** is $\mathring{\Delta}^n := \Delta^n - \delta\Delta^n$.

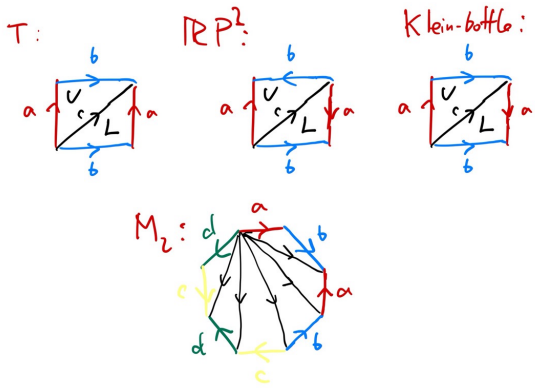
Such simplexes are used to define both simplicial and singular homology. For simplicial homology specifically, we require the following.

Definition 3.1.2: Δ-complex

A **Δ-complex** structure on a topological space, is defined as a collection of maps $\sigma_\alpha : \Delta \to X$ such that the restriction of each map $\sigma_\alpha : \mathring{\Delta}^n \to X$ are injective and non-overlapping (i.e. each $x \in X$ is in the image of exactly one map), the restriction of each $\sigma_\alpha : \Delta^n \to X$ to one of the faces is one of the other maps $\sigma_\beta : \Delta^{n-1} \to X$, and finally, such that $A \subset X$ is open iff. $\sigma_\alpha^{-1}(A)$ is open.

Every Δ-complex is clearly Haussdorff by the last condition. One can even show (I suppose this is quite intuitive...) that they are CW-complexes, so they are not very generic, but they provide good intuition anyway for what is coming.

The following figure shows three simple examples of Delta-complexes, that provide for important calculations later



The bottom example is the orientable surface of genus 2 ("2-holed donut"). It generalizes to genus g, by creating a regular $4g$-gon, and identifying the edges according to the word $a_1 b_1 a_1^{-1} b_1^{-1} \dots a_g b_g a_g^{-1} b_g^{-1}$.

Let $\Delta^n(X)$ be the free abelian group with generators all of the (interiors of the images of) n-simplecies. In Hatcher, these images are written as $e_\alpha^n := \sigma_\alpha(\mathring{\Delta}^n)$, i.e. so that $\Delta^n(X) := \bigoplus_\alpha \mathbb{Z} e_\alpha^n$. The boundary map is defined (on simplecies) by sending it to the alternating sum of its faces (the

signs are philosophically supposed to take the orientation into account)

$$\delta_n : \Delta^n(X) \to \Delta^{n-1}(X)$$
$$\delta_n(\sigma_\alpha) = \sum_{i=1}^{n} (-1)^i \sigma_\alpha \mid_{[v_0,\ldots,\widehat{v_i},\ldots,v_n]} .$$

By direct computation, it one can see that $\delta_{n-1}\delta_n$ is always zero, and thus defines

Definition 3.1.3: Simplicial homology

Singular homology is defined as the homology groups of the chain complex $\Delta^\bullet(X)$, i.e.

$$H_n^\Delta(X) = \ker \delta_n / \operatorname{im} \delta_{n+1}.$$

We can now compute some singular homology groups directly from the definition.

Example 3.1.4

For all three examples in the figure from before, $T, \mathbb{RP}^2, K$ (K for Klein-bottle), we have only three non-trivial Δ^n:

$$\cdots \to 0 \to \Delta^2 \to \Delta^1 \to \Delta^0 \to 0$$

In all three cases, the 1-simplexes are labeled a, b, c, and 2-simplexes U, L. The number of vertecies is not the same: we have the same situation for T and K (though the maps themselves of course differ):

$$\cdots \to 0 \to \mathbb{Z}^2 \to \mathbb{Z}^3 \to \mathbb{Z} \to 0$$

while $\mathbb{RP}^2$ is built from two 0-simplexes (vertecies), so the situation is

$$\cdots \to 0 \to \mathbb{Z}^2 \to \mathbb{Z}^3 \to \mathbb{Z}^2 \to 0$$

From explicitly computing the boundary-maps, we can also compute homology groups. We start with T: Clearly, $\delta_1 = 0$, so we only really need to figure out δ_2. Choosing an orientation on U and V (for instance according to the direction of c, we get $\delta_2 U = c - b - a = \delta_2 L$. Choosing the basis for $\Delta^1(X)$ as $[a, b, c - a - b]$, the homology groups become obvious:

$$H_k^\Delta(T) = \begin{cases} \mathbb{Z} & \text{if } k = 0, 2 \\ \mathbb{Z}^2 & \text{if } k = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Next, for K, we again only need to figure out δ_2. We see that in this case, $\delta_2 U = c - b - a$, while $\delta_2 L = c + a - b$. These are clearly linearly independent, and non-zero, so H^2 must vanish. Choosing the basis $[a, b, c]$, we have $H^1(K) = \mathbb{Z}^3/\Lambda$, where Λ is the lattice spanned with basis

$$\begin{pmatrix} -1 & -1 & 1 \\ 1 & -1 & 1 \end{pmatrix}$$

from which the SNF gives the homology groups:

$$H_k^\Delta(K) = \begin{cases} \mathbb{Z} & \text{if } k = 0 \\ \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} & \text{if } k = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Finally, for $\mathbb{RP}^2$, lets call the vertices v, w. We have $\delta_1 a = \delta_1 b = \delta_1 c = v - w$, which gives H^0. Further, $\delta_1 U = c + b - c$, while $\delta_1 L = c + a - b$ which is again injective. Putting the maps together, it is not hard to see

$$H_k^\Delta(\mathbb{RP}^2) = \begin{cases} \mathbb{Z} & \text{if } k = 0 \\ \mathbb{Z}/2\mathbb{Z} & \text{if } k = 1 \\ 0 & \text{otherwise.} \end{cases}$$

4 Exercise Solutions

Exercise 4.0.A. For a closed orientable surface M of genus $g \geq 1$, show that for each nonzero $\alpha \in H^1(M;\mathbb{Z})$ there exists $\beta \in H^1(M;\mathbb{Z})$ with $\alpha\beta \neq 0$. Deduce that M is not homotopy equivalent to a wedge sum $X \vee Y$ of CW complexes with nontrivial reduced homology. Do the same for closed nonorientable surfaces using cohomology with $\mathbb{Z}/2\mathbb{Z}$ coefficients.

Solution.

□